UNIVERSITY OF TORONTO
Faculty of Arts & Science

August 2023 Examinations

CSC 311 H1Y

Duration: 3 hours

Aids Allowed: One *double*-sided *hand*written *8.5"×11"* aid sheet.

*Do **not** turn this page until you have received the signal to start.*
In the meantime, write your name, student number, and UTORid below (please do this now!) and *carefully* read *all* the information on the rest of this page.

Given Name(s):

Family Name(s):

Student Number (9 or 10 digits):

UTORid (e.g., `pitfra12`):

- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. *Do not collect your bag and jacket before your exam is handed in.*
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- In the event of a fire alarm, do **not** check your cell phone when escorted outside.
- This final examination consists of 5 question on 18 page (including this one), printed on both sides of the paper. *When you receive the signal to start, please make sure that your copy of the examination is complete.*
- Answer each question directly on the examination paper, in the space provided, and use a "blank" page for rough work. If you need more space for one of your solutions, use one of the "blank" pages and *indicate clearly the part of your work that should be marked.*
- *Remember that, in order to pass the course, you must achieve a grade of at least **40%** on this final examination.*
- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.

Marking Guide

Nº 1:	_____/41
Nº 2:	_____/ 6
Nº 3:	_____/ 6
Nº 4:	_____/ 5
Nº 5:	_____/ 6
TOTAL:	_____/64

STUDENTS MUST HAND IN ALL EXAMINATION MATERIALS AT THE END

Question 1. Conceptual Questions: [41 marks]

Answer each question in short sentences.

Part (a) [2 marks]

What is *underfitting* in Machine Learning and how do we prevent it?

Part (b) [5 marks]

What is *overfitting* in Machine Learning? List four methods to prevent overfitting and explain when each method should be preferably used.

Part (c) [2 marks]

Why is it called *Logistic Regression* if it is a classification model? What are the outputs of a logistic regression model?

Part (d) [4 marks]

Explain the *Gradient Descent* algorithm. Write down the update formula in each iteration. Visualize and explain the effects of high and low learning rates.

Part (e) [4 marks]

What are the most common activation functions for the hidden and output layers of a Neural Network. For the output layer, name the activation functions used for each model type (i.e. regression, multi-class classification, and multi-label classification). Write a short sentence for the reason behind using each function.

Part (f) [2 marks]

Explain *Auto-Differentiation* and its relationship with the Backpropagation algorithm.

Part (g) [2 marks]

What are the benefits of *Convolutional Networks* versus *Fully Connected Networks* when we are dealing with images as the model input? List two benefits.

Part (h) [4 marks]

Explain *Equivariance* and *Invariance* properties of Convolutional Networks and their relationship with each layer type of the network (i.e. convolution layer and pooling layer).

Part (i) [3 marks]

Explain the difference between *Discriminative* and *Generative* classifiers. Mathematically explain how the classification task is done in a generative classifier.

Part (j) [2 marks]

What is the Naive assumption in the *Naive Bayes Models*. At what stage will you use Bayes rule in this algorithm?

Part (k) [3 marks]

Explain *Maximum Likelihood Estimation* and *Maximum A-Posteriori (MAP) Estimation* methods. What assumption do we make for MLE that makes it different from MAP.

Part (l) [3 marks]

Explain the two steps of the *Expectation-Maximization* algorithm. When do we use the EM algorithm instead of Gradient Descent or MLE algorithms?

Part (m) [2 marks]

Explain the *Local Minima* issue in the *K-Means* clustering algorithm. Visualize the problem with a simple example.

Part (n) [3 marks]

Explain, in words, the *Exploration-exploitation tradeoff* in Reinforcement Learning. Provide one example of Exploration-Exploitation in the real world.

Question 2. [6 marks]

Suppose we are interested in using PCA to reduce the number of features of a 3-dimensional dataset. The empirical mean of the data is $\hat{\mu} = \begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$, and the empirical covariance matrix has the following spectral decomposition:

$$\hat{\Sigma} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \end{bmatrix}^\top$$

a (2 marks) What is the first (top) principal component? Give your answer as a vector.

b (4 marks) Consider the vector $x = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$. Determine the reconstruction $\tilde{x}$ when projecting x onto the principal subspace spanned by the first principal component. Give your answer as a vector.

Question 3. [6 marks]

Suppose we are building an image classification model with 10 classes. The input images are RGB images with 32×32 pixels. Calculate the number of parameters in each of the following networks. You don't need to simplify your calculations.

a (2 marks) A fully connected network with two hidden layers of 256 and 64 neurons.

b (4 marks) A convolutional neural networks with the following architecture:

(a) a 2D convolutional layer with 32 5×5 filters and stride of 2
(b) a 2×2 max pooling layer
(c) a 2D convolutional layer with 64 3×3 filters and stride of 1
(d) a dense (fully connected) layer of 64 neurons
(e) the output layer

Question 4. [5 marks]

Recall the following MDP from lecture, where an agent is trying to reach the goal, actions correspond to the four coordinate directions, and each action incurs a reward of -1. According to our analysis in lecture, in the undiscounted setting, the state s shown with a $*$ has a value of $V^\pi(s) = -3$. Suppose that instead we use a discounted return with discount factor $\gamma = \frac{1}{2}$. Determine $V^\pi(s)$.

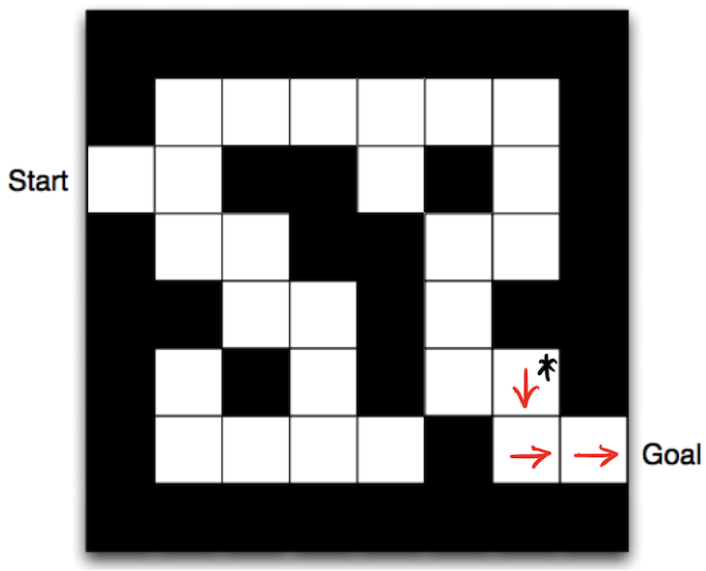

Figure 1: Maze problem.

Question 5. [6 marks]

You are trying to solve a binary classification problem with inputs x and targets $t \in \{0, 1\}$. The data is not balanced and the number of samples in the class $t = 1$ is twice of the number of samples in the class $t = 0$.

$$
\begin{aligned}
x \sim & \mathcal{N}(0, 1) \quad \text{if } t = 0; \\
x \sim & \mathcal{N}(3, 4) \quad \text{if } t = 1; \\
\mathcal{N}(\mu, \sigma^2) = & \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).
\end{aligned}
$$

a (3 marks) Drive $p(t|x)$ for both classes and draw them on one plot (approximately).

b (3 marks) What is the decision boundary for Bayes optimal prediction?

Use the space on this "blank" page for scratch work, or for any solution that did not fit elsewhere.
Clearly label each such solution with the appropriate question and part number.

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